

AI61201: Visual Computing with AI/ML

Programming Assignment 3: Frequency Domain Analysis (20 Marks)

Due Date: **September 16** (by **9 PM** IST)

Instructions: Complete the three tasks given, keeping the following points in mind:

- Implement the Fourier transform, shifting and inverse Fourier Transform operations from scratch (without using OpenCV or NumPy).
- Vectorized implementations are possible for most operations, and using for or while loops to access individual pixels is strongly discouraged due to its inefficiency. You can use NumPy to perform the mathematical operations, except Fourier transform, shifting and inverse Fourier Transform.

Task 1: Effect of low and high frequencies on image reconstruction error (2 + 3 + 1 + 2 = 8 Marks)

- a. Perform the DFT of the given image from scratch (a_3_1.png)
- b. Perform inverse DFT (IDFT) from scratch by successively thresholding the DFT coefficients such that the coefficients (both real and imaginary parts) are set to 0 beyond a certain radial cut-off frequency, i.e. if $\sqrt{u^2 + v^2} > D_0$ then $F(u, v) = 0$ where $F(u, v)$ is the DFT of the image $f(x, y)$.
- c. Compute the reconstruction error between the original image and the reconstructed image in terms of MSE by varying the threshold D_0
- d. Plot the reconstructed error as a function of the frequency threshold D_0 and explain the nature of the plot you obtained.

Task 2: Creating a hybrid image in the frequency domain (1 + 1 + 2 + 1 = 6 Marks)

- a. Perform low pass Gaussian filtering in the frequency domain on the given cat image (a_3_2_1.png). Take the cutoff (standard deviation) at 0.015 times the width (the width and height of the cat image are both 512).
- b. Perform high pass Gaussian filtering on the given dog image (a_3_2_2.png). Take the cutoff (standard deviation) at 0.015 times the width (the width and height of the dog image are both 512).
- c. Create a hybrid spectrum by combining the filtered spectra of the dog and cat images (by summing the DFTs coefficients of the filtered images)
- d. Perform IDFT using the hybrid spectrum and show the image obtained after IDFT.
- e. Is there any effect of the viewing distance on the type of animal (dog or cat) that you perceive? Explain your observation.

Task 3: Watermarking in the frequency domain (1 × 6 = 6 Marks)

- a. Perform the DFT of the given base image (a_3_1.png) and display the magnitude and phase components of the centered DFT.
- b. Embed a small watermark image (a_3_watermark.png) in the mid-band of frequencies by replacing the values of the magnitude spectrum with the pixels of the watermark image. Your embedding operation must ensure that the even symmetry of the magnitude spectrum is preserved. NOTE: Scale the watermark image amplitude to the range of [0.0, 50.0] before embedding, so that it is visible in the magnitude plot.
- c. Show the modified magnitude spectrum.
- d. Using the modified magnitude spectrum and the original phase spectrum perform IDFT to reconstruct the image.
- e. Submit the reconstructed image as a .png file.
- f. What would happen if you would have embedded the watermark at very low or very high frequency regions of the magnitude spectrum? Explain.

Link to download the inputs for all tasks

https://drive.google.com/file/d/1KPr-x5Gr3faWARmvqQYwy5CRZsCpXGn-/view?usp=s_haring

Submission Guidelines

1. *The content that you submit must be your individual work.*
2. *Submit your code in .py as well as in .ipynb file format. Both these file submissions are required to receive credit for this assignment*
3. *Ensure your code is well-commented and easy to follow. You can write your answers and explanations using text cells in the jupyter notebook files wherever required. For example, in Task 1 (d) and Task 3 (f), you need to explain your answer using text.*
4. *Submit output files wherever instructed. For example, in Task 3 (e), it is needed to submit the output image as a .png file.*
5. *The files should be named as "<roll_number>_assignment_3". For example, if your roll number is 23AI91R01, the code the required file names will be 23AI91R01_assignment_3.py, 23AI91R01_assignment_3.ipynb and 23AI91R01_Task3e_output.png. You should place all these files within a single zip file and upload it to Moodle as 23AI91R01_assignment_3.zip.*
6. *All submissions must be made through Moodle before the deadline. The submission portal will close at the specified time, and submissions via email will not be accepted.*

7. *If you have any queries regarding Assignment 3, please email at rajkrishanghosh@gmail.com*